VISVESVARAYA TECHNOLOGICAL UNIVERSITY

JANA SANGAMA, BELAGAVI - 590014

Mini Project Report on

SCREENING AND EVALUATION OF PHYTOCONSTITUENTS OF *Lithospermum officinale* FOR THYROID DRUG TARGETS

Submitted in the partial fulfillment of the requirements for the award of the Degree of

Bachelor of Engineering

In

Biotechnology

Submitted by

Gagan KC(1OX21BT009)
Govindaraju HN(1OX21BT010)
Harini R(1OX21BT011)
Kowshalya C(1OX21BT015)

Under the support and guidance of

Project Guide
Dr. Amulya Giridasapapa
Assistant Professor
Dept. of Biotechnology

Head of the Department
Dr. B K Manjunatha
Professor & Head
Dept. of Biotechnology

CHILDREN'S EDUCATION SOCIETY(REGD)

THE OXFORD COLLEGE OF ENGINEERING

Department of Biotechnology
Bommanahalli, hosur road, Bangalore – 560068
(Approved by AICTE, Accredited by NBA and NAAC with A grade, New Delhi & Affiliated to VTU, Belgaum)
080 – 30219601/2, Fax: 080 – 25730551,30219629,
Website: www.theoxford.edu

THE OXFORD COLLEGE OF ENGINEERING
BOMMANAHALLI, HOSUR ROAD BANGALORE- 560068

(Affiliated to VTU, Belagavi, Approved by AICTE& Accredited by NAAC & NBA ,New Delhi)

DEPARTMENT OF BIOTECHNOLOGY

CERTIFICATE

This is to certify that the project work entitled **SCREENING AND EVALUATION OF PHOTO-CONSTITUENTS OF *LITHOSPERMUM OFFICINALE* FOR THYROID DRUG TARGETS** carried out by **Gagan KC(1OX21BT009)**, **Govindaraju HN(1OX21BT010), Harini R(1OX21BT011), Kowshalya C(1OX21BT015)** bonafide student of **THE OXFORD COLLEGE OF ENGINEERING**, in partial fulfillment to the award of Bachelor of Engineering in Biotechnology of the Visveswaraya Technological University, Belagavi during the year 2023. It is certified that all corrections/suggestions indicated for Internal Assessment have been incorporated in the report deposited in the departmental library. The project report has been approved as it satisfies the academic requirements in respect of project work prescribed for the said Degree.

Signature of the Guide	Signature of the HOD	Signature of the Principal
Dr.Amulya Giridasapapa	**Dr. B K Manjunatha**	**Dr. H N Ramesh**

DECLARATION

We, **Gagan KC(1OX20BT009),Govindaraju HN(1OX21BT010), Harini R(1OX21BT011), Kowshalya C(1OX21BT015)** do hereby declare that the project work entitled **SCREENING AND EVALUATION OF PHYTOCONSTITUENTS OF *Lithospermum officinale* FOR THYROID DRUG TARGETS** is submitted in the partial fulfillment for the award of the degree of Bachelor of Engineering in Biotechnology Engineering, is the result of the research work carried out in the Department of Biotechnology, The Oxford College of Engineering, Bangalore.

I further declare that, this or no part of this has been previously formed the basis of the award of any degree, Associate ship, Fellowship etc., of other University or Institution.

NAME OF STUDENT.	USN	SIGNATURE
GAGAN KC	**(1OX21BT009)**	
GOVINDARAJU HN	**(1OX21BT010)**	
HARINI R	**(1OX21BT011)**	
KOWSHALYA C	**(1OX21BT015)**	

PLACE: Bangalore

DATE:

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ABSTRACT

The study is based on docking tests of *Lithospermum officinale* plant constituents on multiple thyroid drug targets. Utilizing a program called Auto dock Vina, which predicts the ligand's binding to the target protein, molecular docking studies were conducted. Predicting the binding mode(s) of a ligand with a protein having a known three-dimensional structure is the main aim of ligand-protein docking. The ligand lithospermic acid from *Lithospermum officinale* was docked with the target thyroid-stimulating hormone receptor proteins, which were thyroid receptor proteins. Depending on how prevalent the thyroid kind is, specific therapeutic targets were selected. However, in the ligand molecules that were selected for the experiments, a few of them have already demonstrated anti-thyroid action against a particular thyroid hormone. The docking studies were carried out, and the outcomes were analysed and compared with visualization tools such as Pymol.

Keywords: *Lithospermum officinale*; phytoconstituents; Auto dock; Thyroid; ligand-protein docking.

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CHAPTER 1
INTRODUCTION

1.1 INTRODUCTION

- The thyroid, or thyroid gland, is an endocrine gland in vertebrates. In humans, it is in the neck and consists of two connected lobes. The lower two thirds of the lobes are connected by a thin band of tissue called the isthmus (pl.: isthmus). The thyroid gland is a butterfly-shaped gland located in the neck below the Adam's apple.
- The thyroid gland secretes three hormones: the two thyroid hormones – triiodothyronine (T3) and thyroxine (T4) – and a peptide hormone, calcitonin. The thyroid hormones influence the metabolic rate and protein synthesis and growth and development.
- Secretion of the two thyroid hormones is regulated by thyroid-stimulating hormone (TSH), which is secreted from the anterior pituitary gland. TSH is regulated by thyrotropin-releasing hormone (TRH), which is produced by the hypothalamus.
- Thyroid disorders include hyperthyroidism, hypothyroidism, thyroid inflammation (thyroiditis), thyroid enlargement (goitre), thyroid nodules, and thyroid cancer.

Types of Thyroidism

1. Hypothyroidism

Hypothyroidism occurs when the thyroid gland does not produce enough thyroid hormones (thyroxine, T4, and triiodothyronine, T3). This can lead to a slowing down of bodily functions.
Causes:
- Hashimoto's Thyroiditis: An autoimmune disorder where the immune system attacks the thyroid gland.
- Iodine Deficiency: Iodine is essential for thyroid hormone production.
- Thyroid Surgery or Radiation Therapy: Removal or damage to the thyroid can reduce hormone production.
- Medications: Certain medications can affect thyroid function.

2. Hyperthyroidism

Hyperthyroidism is characterized by the overproduction of thyroid hormones, leading to an acceleration of metabolic processes.

Causes:
1.Thyroid Nodules: Autonomous thyroid nodules or toxic adenomas that produce excess thyroid hormone.
2. Thyroiditis: Inflammation of the thyroid gland, which can cause the gland to leak stored thyroid hormone.
-Subacute thyroiditis: Often follows a viral infection.

- Postpartum thyroiditis: Occurs after childbirth.
- Silent thyroiditis: Painless and occurs with or without a known cause.

3. Excess Iodine: High levels of iodine can lead to overproduction of thyroid hormone. This is found in medications, supplements, and iodine-rich foods.
4. Excessive Thyroid Hormone Medication: Taking too much thyroid hormone medication can lead to hyperthyroidism.

5. Toxic Multinodular Goiter: Multiple nodules in the thyroid that produce excess thyroid hormone.
6. Functioning Thyroid Cancer Metastasis: Rarely, thyroid cancer can produce excessive thyroid hormone.
7. Pituitary Adenomas: Rare tumors of the pituitary gland that can stimulate the thyroid gland to produce excess hormone.

Symptoms:
- Weight loss
- Rapid or irregular heartbeat
- Increased appetite
- Nervousness or anxiety
- Sweating
- Heat intolerance
- Frequent bowel movements

Diagnosis and Treatment:
Diagnosis involves blood tests for TSH and thyroid hormone levels, radioactive iodine uptake tests, and thyroid scans. Treatment options include anti-thyroid medications, radioactive iodine therapy, and surgery.

Graves' Disease
Graves' Disease is an autoimmune disorder and the leading cause of hyperthyroidism. It results from the immune system producing antibodies called thyroid-stimulating immunoglobulins (TSI) that bind to thyroid receptors, stimulating the gland to produce excessive thyroid hormones.

Causes
- Genetics: Family history increases risk.
- Gender: More common in women.
- Age: Typically diagnosed in people younger than 40.
- Other Autoimmune Disorders: People with other autoimmune diseases, such as rheumatoid arthritis or type 1 diabetes, are at higher risk.

Symptoms:

- Feeling nervous and irritable.
- Having a slight tremor of the hands or fingers.
- Being sensitive to heat with an increase in sweating or warm, moist skin.
- Losing weight, despite wanting to eat more.
- Having an enlarged thyroid gland, also called goiter.

- Having changes in menstrual cycles.
- Not being able to get or keep an erection, called erectile dysfunction, or
- having less desire for sex.
- Having bowel movements often.
- Being tired.
- Having thick, discolored skin mostly on the shins or tops of the feet, called Graves' dermopathy.
- Having fast or irregular heartbeat, called palpitations.

Diagnosis:
- Blood Tests: Measure levels of TSH, T3, and T4, and test for TSI antibodies.
- Radioactive Iodine Uptake: Measures how much iodine the thyroid absorbs.
- Ultrasound or Thyroid Scan: Visualize the thyroid gland's structure and activity

Treatment:
-Antithyroid Medications: Such as methim
-Medications: Anti-thyroid medications such as methimazole and propylthiouracil can reduce thyroid hormone production.
- Radioactive Iodine Therapy: This treatment destroys overactive thyroid cells.
- Surgery: In some cases, part or all of the thyroid gland may be surgically removed

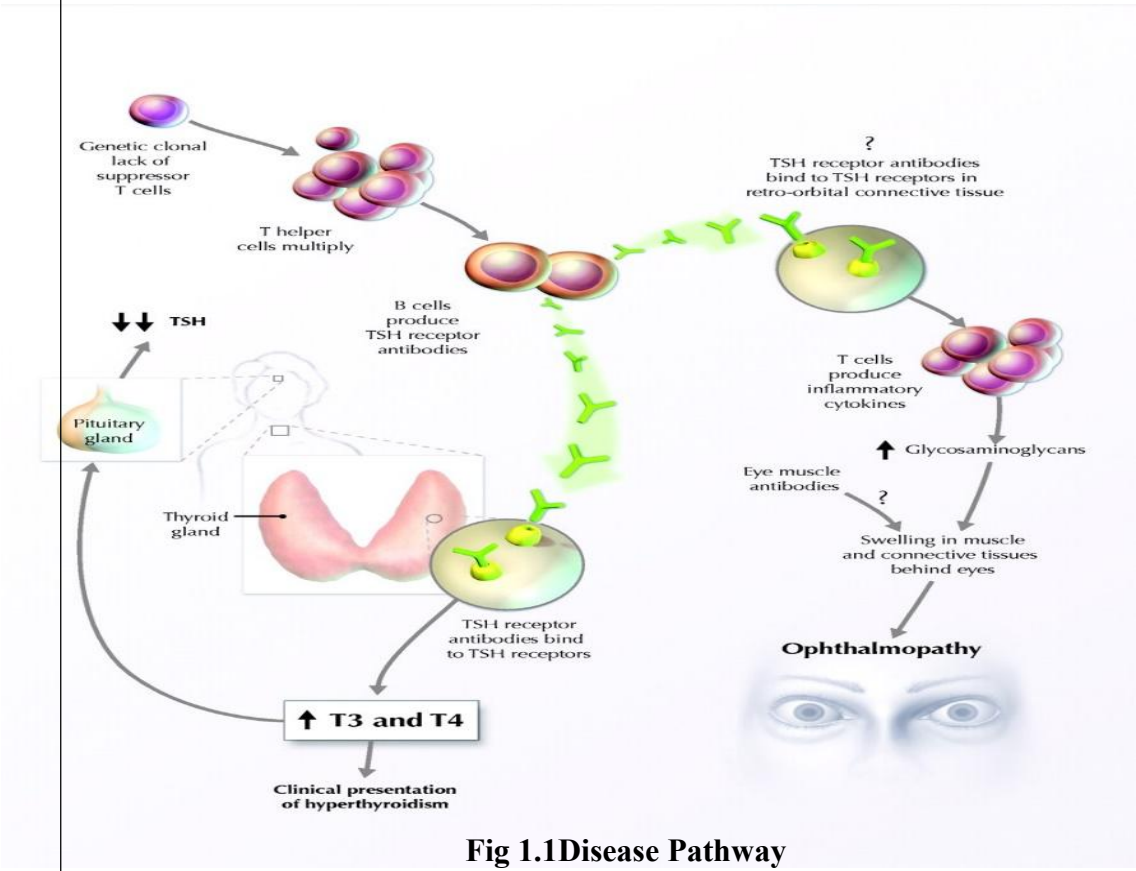

Fig 1.1Disease Pathway

Statistics of Hyperthyroidism

TABLE 1:

The following table gives the comparision of hypothyroidism and hyperthyroidism:

Age Group	Hyperthyroidism	Hypothyroidism	Hyperthyroidism %	Hypothyroidism %
10 to 20	293	879	3.63	10.9
20 to 30	2396	6611	4.44	12.25
30 to 40	1283	5483	4.34	18.56
40 to 50	781	3327	5.33	22.71
50 to 60	562	1942	6.35	21.93
60 to 70	403	1114	7.68	21.22
70 to 80	133	318	7.81	18.68
Above 80	39	93	8.52	20.31
Total	**5890**	**19767**	**4.81**	**16.14**

TABLE 2:
Here is a table summarizing some key statistics hyperthyroidism specifically Grave's disease

SATISTICS	VALUES
1. Prevalence	0.5% of the population
2. Incidence rate	20-30 cases per 100,000 people per year
3. Gender distribution	Women : Men = 7:1
4. Age of onset	20-40 years
5. Familial link	15-20% have a family history
6. Autoantibody presence	90-95% of patients
7. Thyroid eye disease co-occurrence	20-25% of patients
8. Treatment response rate	Antithyroid drugs: 30-50 remission
9. Radioactive iodine therapy	70-90% effectiveness
10. Surgical treatment	90-95% cure rate
11. Relapse rate	50-60% within first year after stopping medication
12. Mortality rate	Less than 1% with treatment

1.5 *Lithospermum officinale*

Figure 2: ***Lithospermum officinale***

Lithospermum officinale, or common gromwell or European stoneseed, is a flowering plant species in the family Boraginaceae, native to Eurasia. It is the host plant for caterpillars of the monophagous moth Ethmia dodecea.
All parts of the plant have been traditionally used as a natural medicinal remedy for various ailments. In India for example, the leaves were once used as a sedative, while the seeds have been administered as a diuretic, lithotritic, febrifuge, anti-gout, anti-ovulary and anti-toxic agent as well as an anti-inflammatory for diseases of the urinary tract and for promoting digestion. A herbal tea made from the root and stem, or a decoction of the roots and twigs was once given in the form of a syrup to remedy smallpox and measles. The tea also served as an antipruritic Archaeological findings from a gravesite in Poland unearthed a plaster poultice containing the nutlets of L. officinale. The plant has been found to be a potent natural anti-inflammatory and effective agent for healing burn wounds when applied topically, which explains the presence of this species in the poultice discovered. L. officinale also appears to have held important cosmetic and ornamental value. For example, the roots were once used for colouring fibres and to produce makeup dye.

***Lithospermum officinale* Overview**

Common Name: Common gromwell
Family: Boraginaceae
Description:
Lithospermum officinale is a perennial herbaceous plant known for its distinctive hard, white, stony seeds. It typically grows to a height of 30-80 cm. The plant has lanceolate leaves that are rough and hairy. It produces small, tubular flowers that are usually white or pale yellow and bloom from May to July. **Habitat and Distribution:**
Lithospermum officinale is native to Europe and parts of Asia. It prefers calcareous soils and is commonly found in grasslands, woodland edges, and scrubby areas. It has also been introduced to North America, where it grows in similar habitats.
Phytochemical Constituents
Lithospermum officinale contains several bioactive compounds, including:

Lithospermum officinale

Scientific classification

Kingdom:	Plantae
Clade:	Tracheophytes
Clade:	Angiosperms
Clade:	Eudicots
Clade:	Asterids
Order:	Boraginales
Family:	Boraginaceae
Genus:	*Lithospermum*
Species:	***L. officinale***

Binomial name

Lithospermum officinale

Fig 3 : Description of plant

- Alkaloids: These are nitrogen-containing compounds that can have significant physiological effects on humans and animals.
- Phenolic acids: Known for their antioxidant properties, these compounds help in protecting cells from oxidative stress.
- Flavonoids: These compounds have anti-inflammatory, antioxidant, and antimicrobial properties.
- Fatty acids: Essential for various bodily functions, they also contribute to the plant's medicinal properties

Medicinal Uses
Lithospermum officinale has a long history in traditional medicine. It has been used for its sedative, diuretic, and anti-inflammatory properties.
Modern research has validated many of the traditional uses of Lithospermum officinale and has also uncovered new potential applications:
1. Anti-Inflammatory and Antioxidant Effects:
   - The plant's phenolic acids and flavonoids contribute to its ability to reduce inflammation and protect against oxidative stress
2. Antimicrobial Properties:
   - The presence of certain bioactive compounds gives Lithospermum officinale antimicrobial properties, making it effective against various bacterial and fungal infections.
3. Antithyroid Activity:
   - Lithospermum officinale is noted for its ability to inhibit thyroid function, which is particularly useful in managing hyperthyroidism and autoimmune thyroid conditions such as Graves' disease. It achieves this by inhibiting the conversion of T4 to T3, blocking iodine uptake, and binding to TSH receptors

1.6 MOLECULAR DOCKING

Molecular docking has grown to be a significant step in the drug discovery process. Since docking was first developed in the 1980s, improvements in computer hardware power, the number of and ease of access to small molecule and protein structures, and the development of better techniques have all contributed to the development of improved methods, making docking more well-liked in both academic and industrial settings. The methods docking is used to support the various tasks of drug discovery have evolved over time. Docking was initially created and used as a stand-alone technique, but it is now primarily used in conjunction with other computational techniques as part of integrated workflows.

Molecular docking is a kind of Bioinformatic modelling which involves the interaction of two or more molecules to give the stable adduct. Depending upon binding properties of ligand and target, it predicts the three-dimensional structure of any complex. Molecular docking generates different possible adduct structures that are ranked and grouped together using scoring function in the software.

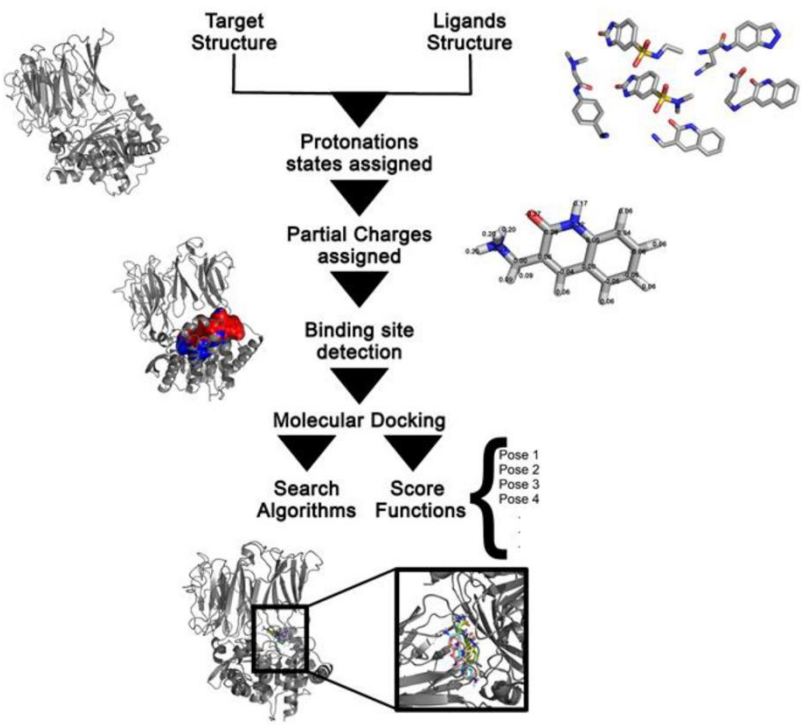

Figure 4: Overview of Molecular Docking

CHAPTER 2
LITERATURE REVIEW

2.1 PURPOSE OF LITERATURE REVIEW

The literature review was done to find the following:

1. Phytochemical constituents of *Lithospermum officinale*
2. Properties of *Lithospermum officinale*
3. Properties of phytochemical constituents of *Lithospermum officinale*
4. Traditional uses of *Lithospermum officinale*
5. To understand about the thyroid selected for the study.
6. To perceive the molecular docking procedure.

2.2 LITERATURE REVIEW

2.2.1 *Lithospermum officinale*

Lithospermum officinale

Aknur Turgumbayeva et.al. (2019) stated that *Lithospermum officinale* L., found in Eurasia, Argentina (South), Colombia, and the United States, is valued for its medicinal properties, including anti-inflammatory, antioxidant, and antimicrobial effects. The current review emphasizes *L. officinale* L. as a significant reservoir of bioactive phytochemicals, with alkaloids, quinones, glucosides, phenolics, flavonoids, and lipids identified as the principal metabolites. It also unveils the unexplored potential of this plant for future research endeavors. Continued research on *L. officinale* L. can unlock its full potential, providing insights into its medicinal uses and contributing to biodiversity preservation.

Olena Andrushchenko et.al. (2017) stated that The features of *Lithospermum officinale* ontogeny during growing in the Forest-steppe zone of Ukraine are that in the first year of vegetation of plants there are all age states – from sprout to the generative age state, which ends with the formation of a small amount of viable seeds. After the cutting of the aboveground part, there was the formation of proleptic vegetative-generative shoots, on which buds, flowers, and several erems were formed. The formation of sylleptic shoots is characteristic of perennial plants in the second decade of March. When cultivating *Lithospermum officinale* in the studied conditions it is recommended a sowing before winter, or preliminary stratification of seeds.

Omid Khanehbarndaz et.al. (2019) stated that The present study was conducted to investigate the molecular identification of endophytic fungi isolated from the roots of Lithospermum *officinale* and their potential production of shikonin. Phylogenetic analysis was performed based on the Internal Transcribed Spacer (ITS) region and the isolates were classified into five genera as follows: Alternaria, *Chaetosphaeronema*, Fusarium, Mucor, and Trichoderma. The study on the methanol extracts of endophytic fungi indicated that total polyphenol content had a positive relationship with antioxidant activities and the highest antioxidant activity belonged to the methanol extracts of Fusarium *tricinctum* and Alternaria *altenata*

Amit Rai et.al. (2018) stated that *Lithospermum officinale* is a valuable source of bioactive metabolites with medicinal and industrial values. However, little is known about genes involved in the biosynthesis of these metabolites, primarily due to the lack of genome or transcriptome resources. This study presents the first effort to establish and characterize *de novo* transcriptome assembly resource for *L. officinale* and expression analysis for three of its tissues, namely leaf, stem, and root. Using over 4Gbps of RNA-sequencing datasets, we obtained *de novo* transcriptome assembly of *L. officinale*, consisting of 77,047 unigenes with assembly N50 value as 1524 bps. Based on transcriptome annotation and functional classification, 52,766 unigenes were assigned with putative genes functions, gene ontology terms, and Kyoto Encyclopedia of Genes and Genomes (KEGG) pathway

J Cohen et.al. (2022) stated that *Lithospermum* is resolved as monophyletic, and the New World species of the genus are also monophyletic. While minimal phylogenetic incongruence is resolved within the nuclear genome, incongruence between the nuclear and plastid genomes is recovered. This is likely due to incomplete lineage sorting during early diversification of the genus in the Americas approximately 7.8 million years ago. At least four shifts to longer corollas are identified throughout *Lithospermum*, and this may be due to selection for hummingbird-pollinated flowers, particularly for species in Mexico and the southwestern United States. In the New World, one clade of species of the genus diversified primarily across the United States and Canada, and another radiated throughout the mountains of Mexico.

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2.2.2 GRAVES DISEASE

Terry F. Davies et.al. (2020) stated that Graves' disease (GD) is an autoimmune disease that primarily affects the thyroid gland. It is the most common cause of hyperthyroidism and occurs at all ages but especially in women of reproductive age. Graves' hyperthyroidism is caused by autoantibodies to the thyroid-stimulating hormone receptor (TSHR) that act as agonists and induce excessive thyroid hormone secretion, releasing the thyroid gland from pituitary control. TSHR autoantibodies also underlie Graves' orbitopathy (GO) and pretibial myxoedema

F.Menconi et.al. (2017) stated that Presumably, the extrathyroidal manifestations of GD are due to autoimmunity against antigens common to the thyroid and other affected organs. Although its exact etiology remains to be completely understood, GD is believed to result from a complex interaction between genetic susceptibility and environmental factors. Clinically, GD is characterized by the manifestations of thyrotoxicosis as well as by its extrathyroidal features when present, the latter making the diagnosis almost unmistakable In the absence of ophthalmopathy, the diagnosis is generally based on the association of hyperthyroidism and usually diffuse goiter confirmed with serum anti-TSHR autoantibodies (TRAbs)

A Antonelli et.al. (2020) stated that the genetic architecture of Graves' disease, we conducted a genome-wide association study in 1,536 individuals with Graves' disease (cases) and 1,516 controls. We further evaluated a group of associated SNPs in a second set of 3,994 cases and 3,510 controls. We confirmed four previously reported loci (in the major histocompatibility complex, *TSHR*, *CTLA4* and *FCRL3*) and identified two new susceptibility loci (the *RNASET2-FGFR1OP-CCR6* region at 6q27 ($P_{combined} = 6.85 \times 10^{-10}$ for rs9355610) and an intergenic region at 4p14 ($P_{combined} = 1.08 \times 10^{-13}$ for rs6832151)). These newly associated SNPs were correlated with the expression levels of *RNASET2* at 6q27, of *CHRNA9* and of a previously uncharacterized gene at 4p14, respectively. Moreover, we identified strong associations of *TSHR* and major histocompatibility complex class II variants with persistently TRAb-positive Graves' disease.
Phytochemical constituents and properties

Grigoris Effraimidis et.al (2014) stated that The number of identified susceptibility genes for AITD is increasing (among others now including *TSHR, TG, HLA, CTLA4, PTPN22, CD40, FCRL3, IL2RA*, and *FOXP3*), but these genes together probably do not explain more than about 10% of the heritability of AITD. As twin studies indicate that genes contribute for 70% of AITD, it follows that there must be many more loci, each of them contributing a little. While the genetic studies have clarified why various autoimmune diseases so often cluster in the same patient, the molecular mechanism of action of these genetic polymorphisms (frequently located in introns) has hardly been explained.

Giulia Di Dalmazi et.al. (2021) stated that The anti-thyroid effects of some plant constituents have been known for some time. Indeed, epidemiological observations have shown the causal association between staple food based on brassicaceae or soybeans and the development of goiter and/or hypothyroidism. Herein, we review

the main plant constituents that interfere with normal thyroid function such as cyanogenic glucosides, polyphenols, phenolic acids, and alkaloids. In detail, we summarize the *in vitro* and *in vivo* studies present in the literature, focusing on the compounds that are more abundant in foods or that are available as dietary supplements. We highlight the mechanism of action of these compounds on thyroid cells by giving a particular emphasis to the experimental studies that can be significant for human health.

Bluck Habibi et.al.(2021) stated that silver nanoparticles were synthesized using Lithospermum officinale root aqueous extract and their antioxidant activity was evaluated in vitro. The results showed that 1 ml of the extract could reduce 9 ml of silver ions (1 mM) to silver nanoparticles by heating the reaction mixture (60 ºC) for 6 hours at pH 7.0. The synthesized silver nanoparticles were detected by UV–Vis spectroscopy, TEM, FT-IR, DLS, and XRD. The synthesized silver nanoparticles spectrum had a maximum peak at 390nm, and TEM analysis indicated spherical particles, higher stability (zeta potential: -15.3 mV) and an average size of 7 nm. The antioxidant activity of the synthesized silver nanoparticles was 0.07 mg/ml compared to L. officinale root aqueous extract (0.142 mg/ml) which indicated higher antioxidant activity. So, it is concluded that the synthesized silver nanoparticles could be considered a clinical therapeutic potential due to its antioxidant property

BC Semwal et.al. (2024) stated that chitosan (CH) and gelatine (GE), have a wide range of uses in biological environments. This is primarily attributed to their biocompatibility and the body's recognition of them being as "self" as opposed to "foreign". These polymers provide a dynamic and cell-friendly interaction environment which is a crucial factor for tissue repair and regeneration. Bone regeneration is a highly intricate multistep process involving a plethora of biomolecules, signalling pathways, and stages in cellular activity.

2.3 OUTCOMES OF LITERATURE REVIEW

After the literature review was performed, the following were understood:

1. Aknur Turgumbayeva et.al. (2019) Olena Andrushchenko et.al. (2017) Omid Khanehbarndaz et.al. (2019) Amit Rai et.al. (2018) J Cohen et.al. (2022) in these articles we have understood about the various phytochemicals , their medicinal properties, and their anatomy.

2. Terry F. Davies et.al. (2020) F.Menconi et.al. (2017) A Antonelli et.al. (2020) numerous causes and diagnosis of the disease have been scientifically understood by us. And also the pathway of the disease along with list of proteins have been known to us

3. Grigoris Effraimidis et.al (2014) Giulia Di Dalmazi et.al. (2021) Bluck Habibi et.al.(2021) *BC Semwal et.al.* (2024) our molecular docking results for current medications were compared with the article information and we understood that our score was almost very close to the article score.

CHAPTER 3
OBJECTIVES

3.1 OBJECTIVES

The objectives of the project after the literature review are:

1. To find the compounds present in Lithospermum officinale which possess antithyroid agents, antioxidant properties.
2. To find drug targets for Grave's disease.
3. To obtain the structure of the compounds and targets.
4. To perform docking using AutoDock Vina 1.5.7 and visualize the docked structure of different poses using pymol.

CHAPTER 4
METHODOLOGY

4.1 IDENTIFICATION OF RECEPTOR PROTEINS

The primary information stored in the PDB archive consists of coordinate files for biological molecules. Using online tools, such as the ones on the RCSB PDB website, we were able to search and explore the information under the PDB header, including information on experimental methods and the chemistry and biology of the protein. After gathering information on the PDB entries, we used visualization programs to view the structure of protein. This gave us the important structural features.

The procedure followed:

1. Open the web browser and type the database address: https://www.rcsb.org/

Figure 5: Protein Data Bank homepage

2.The query “Thyroid stimulating hormone was typed in the search bar.

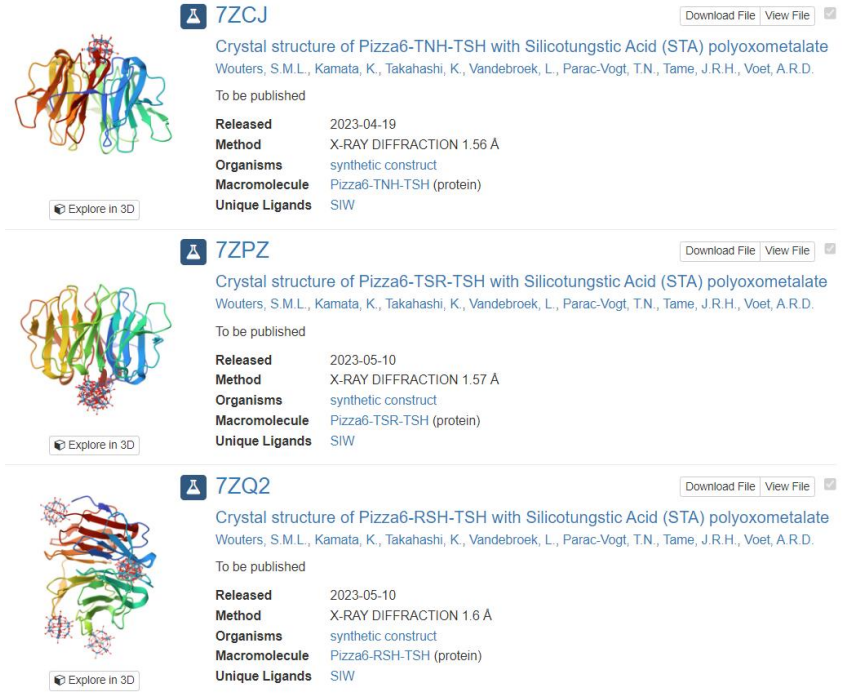

Figure 6: PDB hits

3.The protein of interest was downloaded in pdb format.

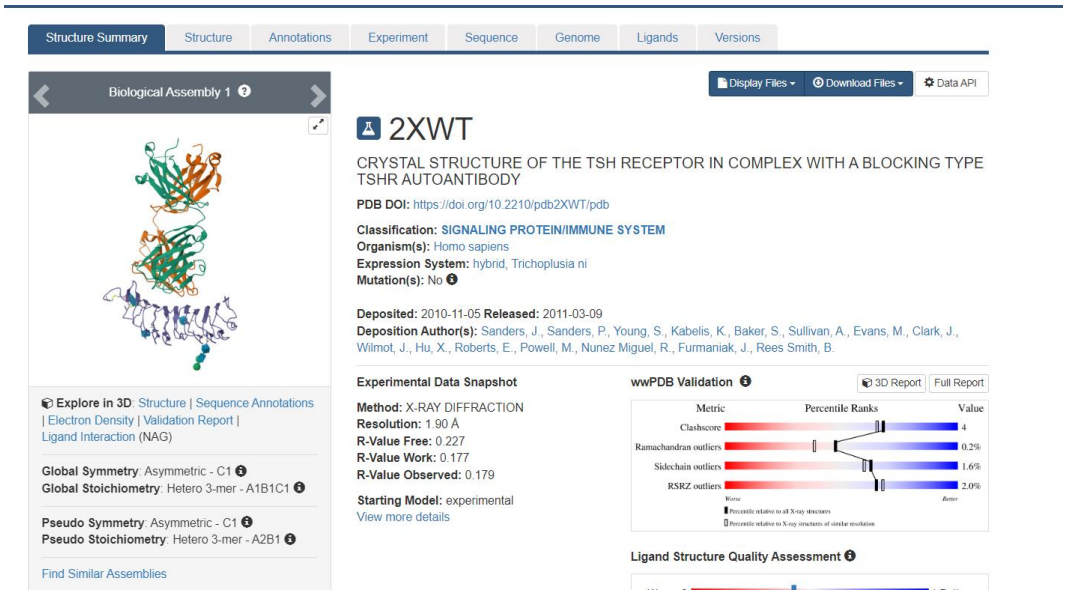

Figure 7: PDB accession ID

4.The following figure is a representation of TSH protein .

Figure 8: 3D structure of TSH protein

4.2 IDENTIFICATION OF LIGAND FROM *Lithospermum officinale*

Following a thorough analysis of the literature review, the ligands isolated from the plant *Lithospermum officinale* are listed [Lithospermic acid, Lithospermine, Lithospermine,Quercetin,Kaempferol ,Isorhapontigenin,Rhamnocitrin,Acteoside,Isoacteoside,Caffeic acid,Ferulic acid,Sinapic acid, Lithospermic acid saponins]. Individual ligands were chosen and their 3D structures were downloaded and examined.

1.Open the web browser and type the database address:

https://pubchem.ncbi.nlm.nih.gov

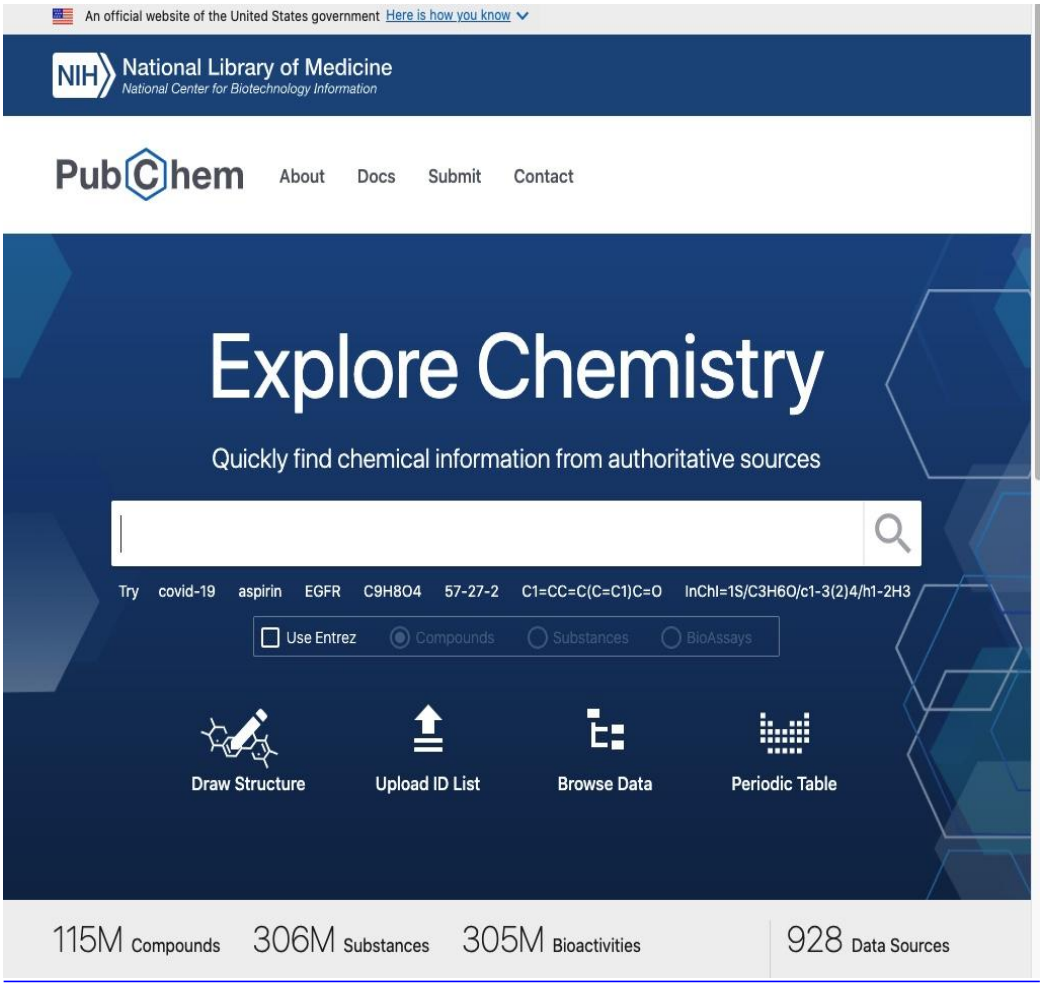

Figure 9: Pubchem database homepage

2.The query ‘Lithospermic acid’ (one of the phytoconstituent of *Lithospermum officinale*) was typed in the search bar.

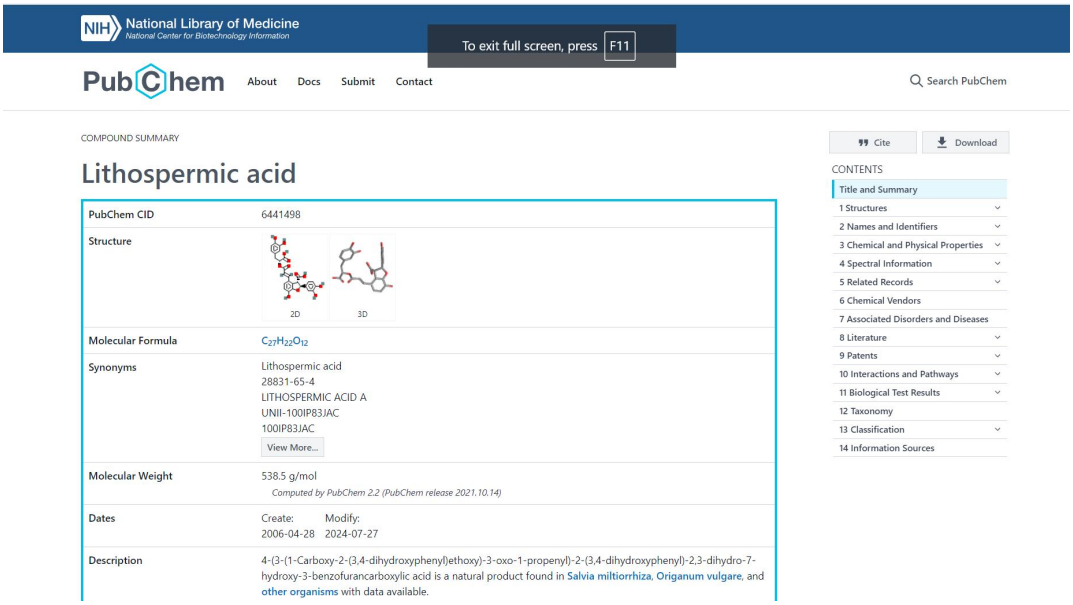

Figure 10: Description of the ligand chosen

3.The structure of ‘Lithospermic acid’ could be vizualized in ball and stick, sticks, wire-frame, space-filling modes. Show hydrogen atoms can also be selected and The co-ordinates of the ligand ‘Lithospermic acid’ was download in sdf format.

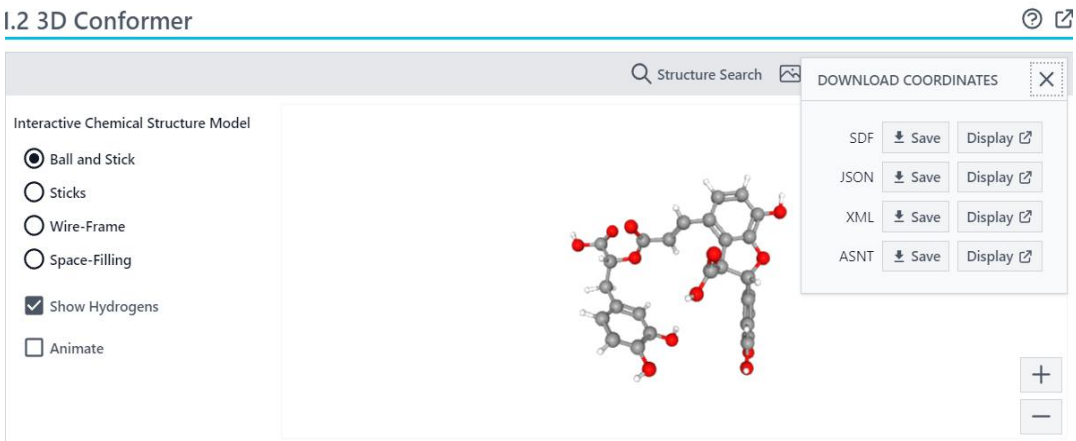

Figure 11: Downloading 3D structure in sdf format

4.3 MOLECULAR DOCKING

Interactions between proteins and peptides are crucial for cell function. Their structural characterization is critical for the identification of novel drugs, despite its difficulty. AutoDock Vina is one of the fastest and most widely used open-source docking engines. It is a turnkey computational docking program that is based on a simple scoring function and rapid gradient-optimization conformational search.

The following procedure is followed:

1. Open the web browser and type the address: https://vina.scripps.edu/
2. Download AUTODOCK vina for Windows Operating system.

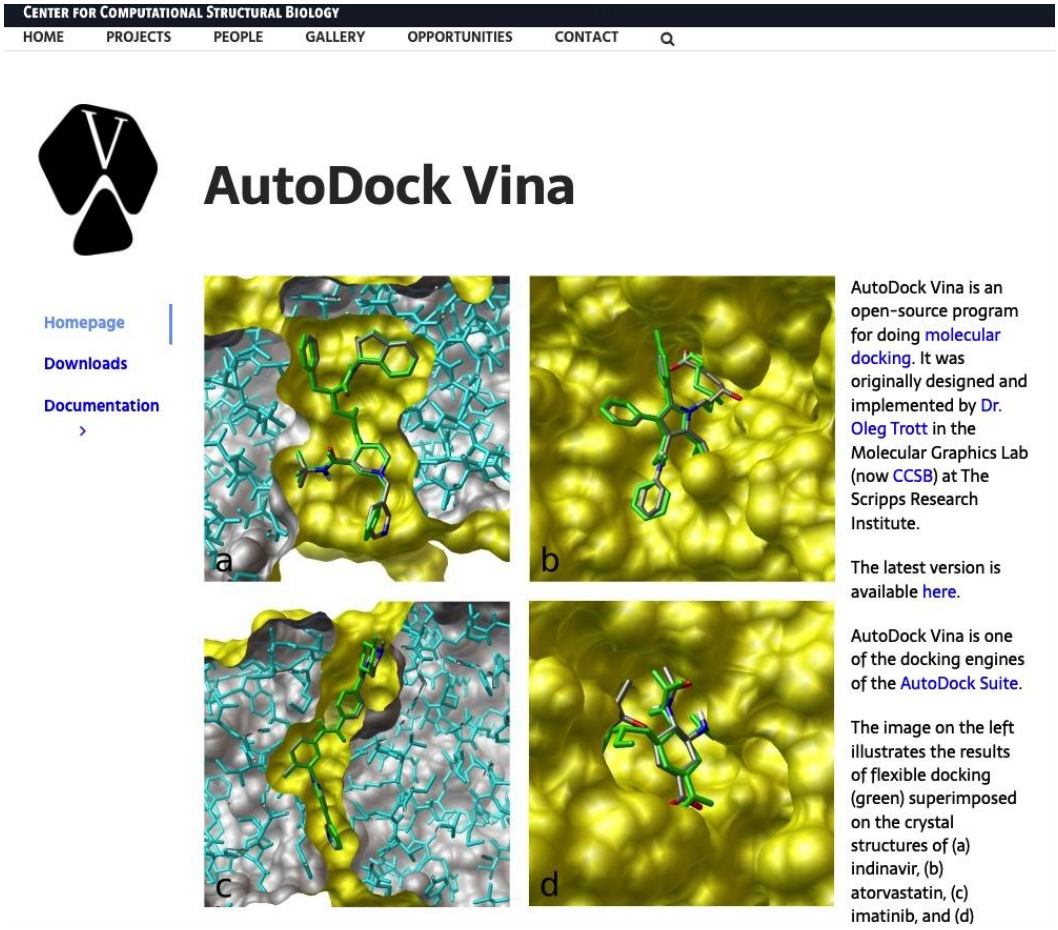

Figure 12.a: Scripps research downloads page

3. Open AUTODOCK vina

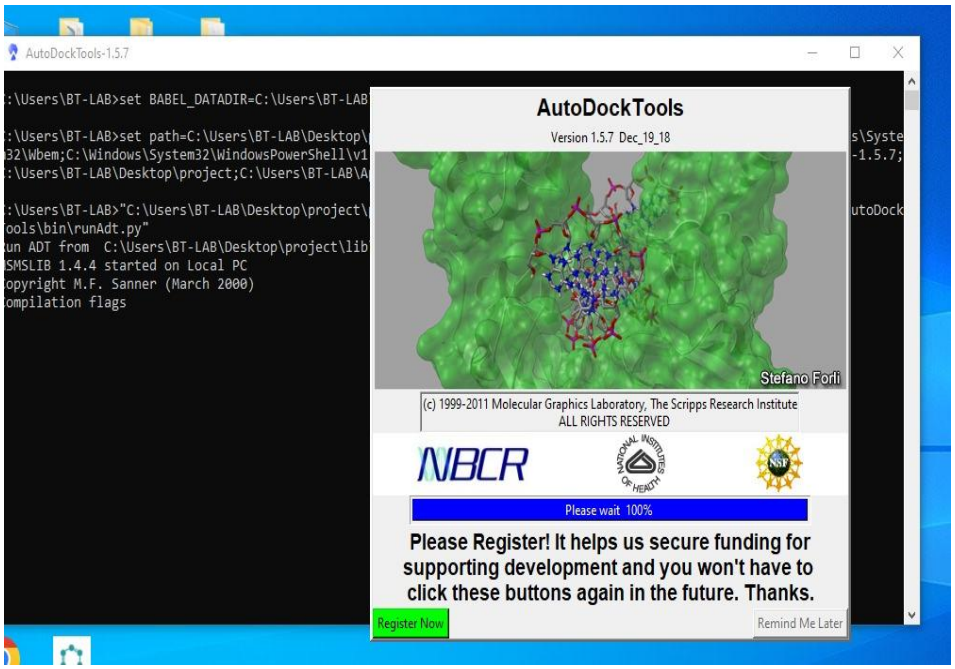

Figure 12.b: AutoDock Vina opening

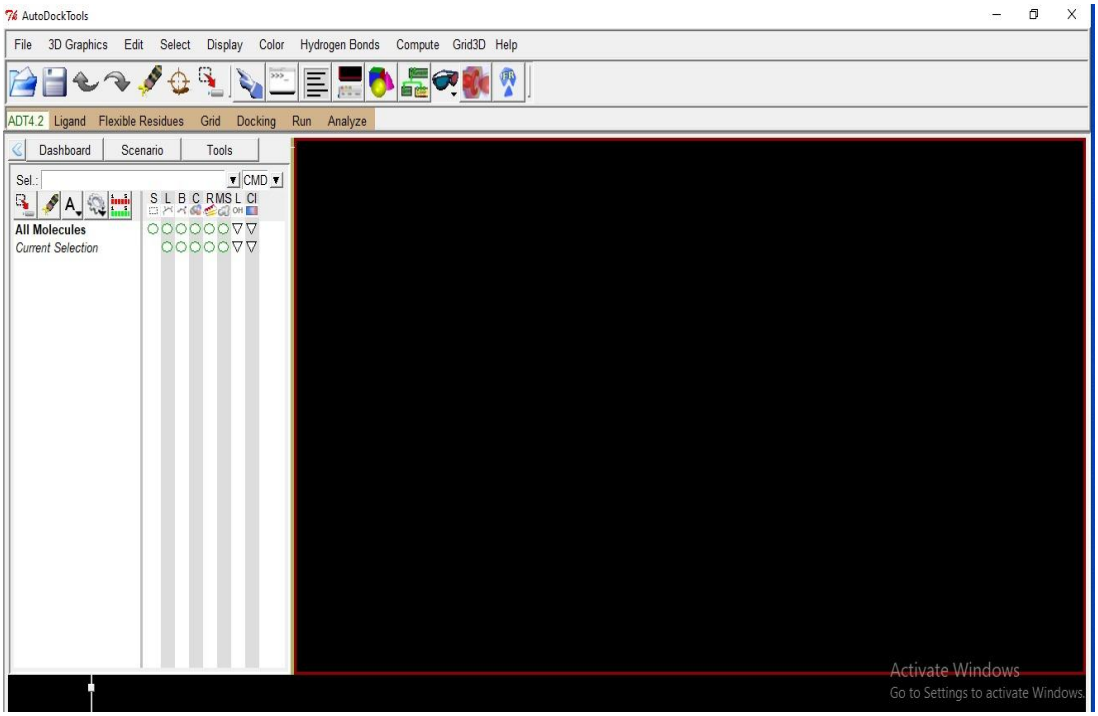

Figure 12.c: AutoDock Vina

4.Select the receptor protein and ligand. The docked pose in AutoDock vina will be observed.

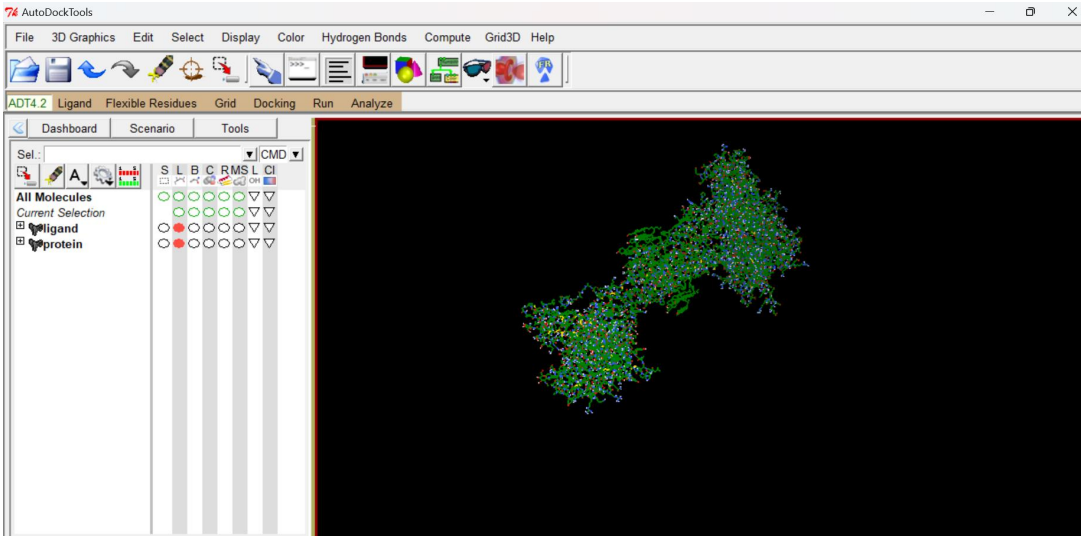

Figure 12.d: Ligand-receptor binding

5.Save the receptor protein and the ligand in .pdbqt format.

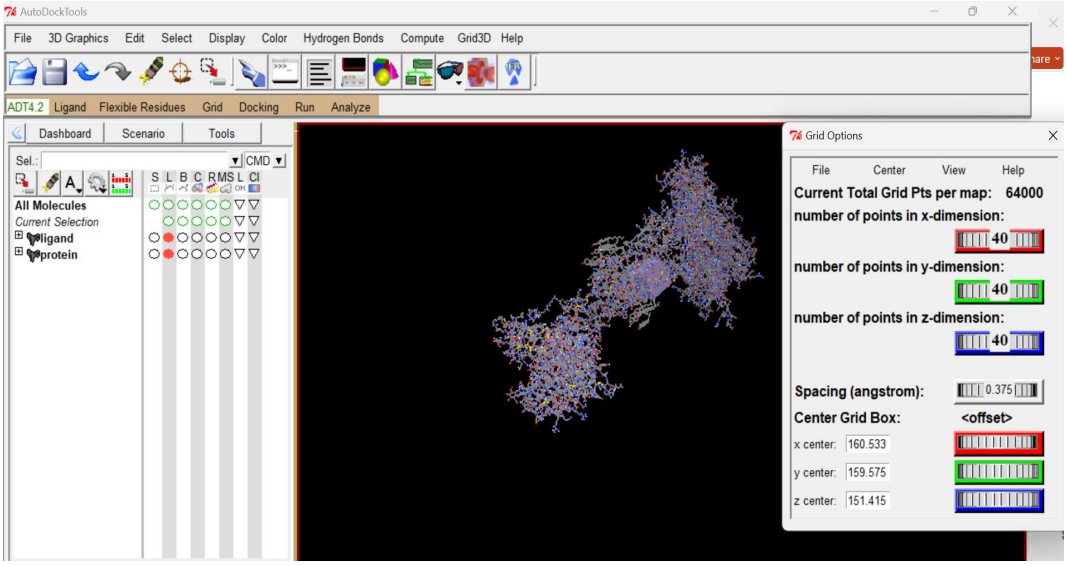

Figure 12.e: Grid box

6.Make a configuration file in .pdbqt format that contains the receptor protein and ligand. The grid box's x, y, and z dimensions are also noted. The exhaustiveness level is set to 8, and the energy range is set to 4. Enter the address of the file containing ligand and receptor in .pdbqt format and file containing the Vina files in the Command Prompt and the output is obtained.

File Edit View

```
protein
spacing    0.375
npts       40 40 40
center     160.533 159.575 151.415
```

File Edit View

```
receptor = protein.pdbqt
ligand = ligand.pdbqt

center_x = 160.533
center_y = 159.575
center_z = 151.415

size_x = 40
size_y = 40
size_z = 40

energy_range = 4

exhaustiveness = 8
```

Figure 12.f: Grid and configuration file

AutoDockTools-1.5.7 | Command Prompt

```
Microsoft Windows [Version 10.0.22631.3880]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Kowshalya c>cd C:\Users\Kowshalya c\Desktop\mp

C:\Users\Kowshalya c\Desktop\mp>"C:\Program Files (x86)\The Scripps Research Institute\Vina\vina.exe" --receptor protein
.pdbqt --ligand ligand.pdbqt --config config.txt --log log.txt --out output.pdbqt
```

Figure 12.g: Command prompt website

CHAPTER 5
RESULTS AND DISCUSSION

Molecular docking studies was done on *Lithospermum officinale* to evaluate it for its anti-thyroid activity by using AutoDock Vina software. The binding energy was obtained and the best ligand-receptor interaction was retained.

RECEPTOR	LIGAND	BINDING ENERGY (kcal/mol)
7UUY (Thyroid Peroxidase)	Lithospermic acid	-8.6
7XW5 (Thyroid stimulating hormone receptor)	Lithospermic acid	-8.2
6SCJ (Thyroglobulin)	Lithospermic acid	-6.9

Table 2: Binding energies of the receptor and ligand

5.1 7UUY AND LITHOSPERMIC ACID

Receptor – 7UUY (Thyroid peroxidase)
Ligand – Lithospermic acid
Binding energy = -8.6kcal/mol

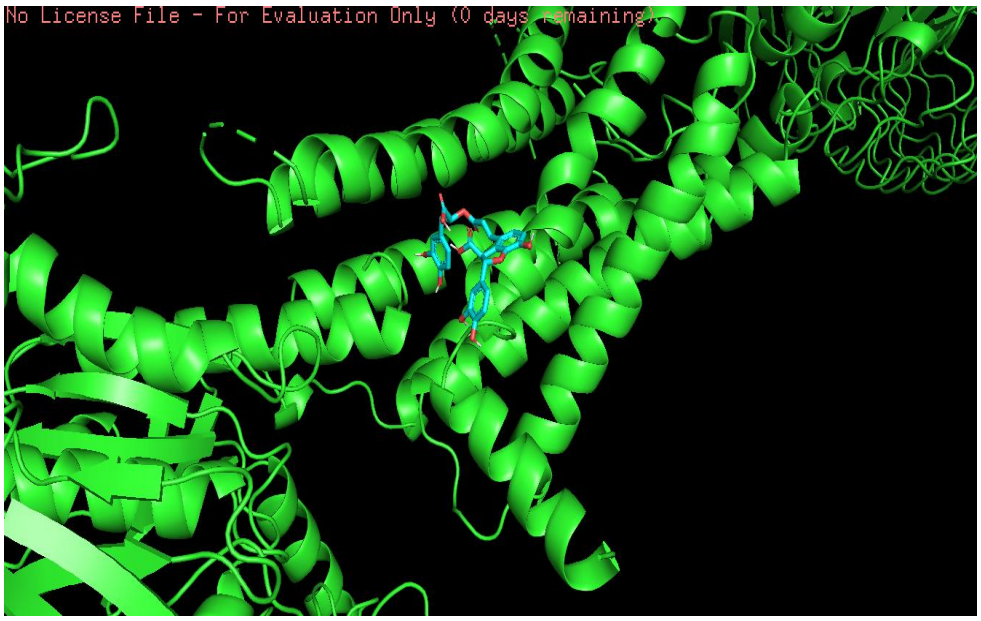

Figure 13 Vizualization of 7UUY and Lithospermic acid in Pymol

5.2 7XW5 AND LITHOSPERMIC ACID

Receptor –7XW5 (TSH)
Ligand – Lithospermic acid
Binding energy = -8.2 kcal/mol

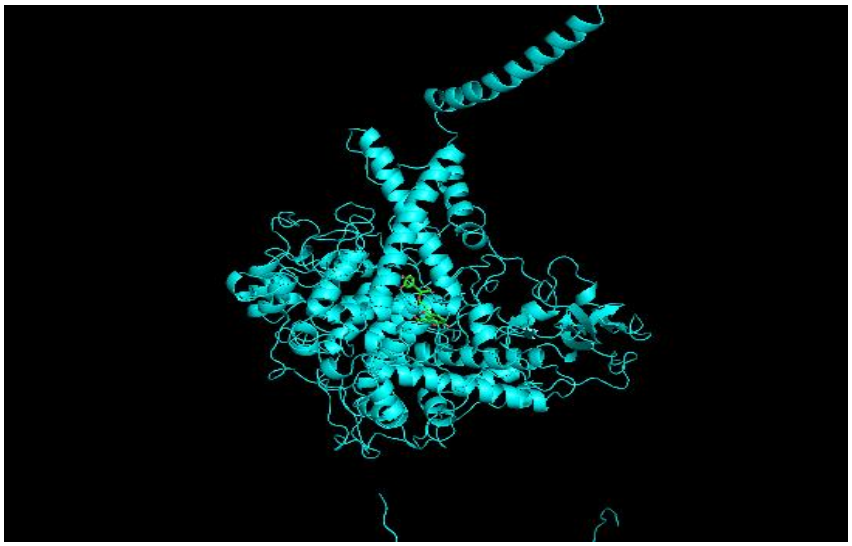

Figure 14: Vizualization of 7XW5 and Lithospermic acid in Pymol

5.3 6SCJ and LITHOSPERMIC ACID

Receptor – 6SCJ (Thyroglobulin)
Ligand – Lithospermic acid
Binding energy = -6.9 kcal/mol

Figure 15: Vizualization of 6SCJ and Lithospermic acid in Pymol

Molecular docking studies was done on FDA approved drug constituents to evaluate it for its anti-thyroid activity by using AutoDock Vina software. The binding energy was obtained and the best ligand-receptor interaction was retained.

RECEPTOR	LIGAND	BINDING ENERGY (kcal/mol)
7UUY (Thyroid Peroxidase)	Methimazole	-4.5
7XW5 (Thyroid stimulating hormone receptor)	Methimazole	-4.3
6SCJ (Thyroglobulin)	Methimazole	-4.0
7UUY (Thyroid Peroxidase)	Propylthiouracil	-3.9
7UUY (Thyroid Peroxidase)	Carbimazole	-3.2

5.4 7UUY AND METHIMAZOLE

Receptor – 7UUY (Thyroid peroxidase)
Ligand –Methimazole
Binding energy = -4,5kcal/mol

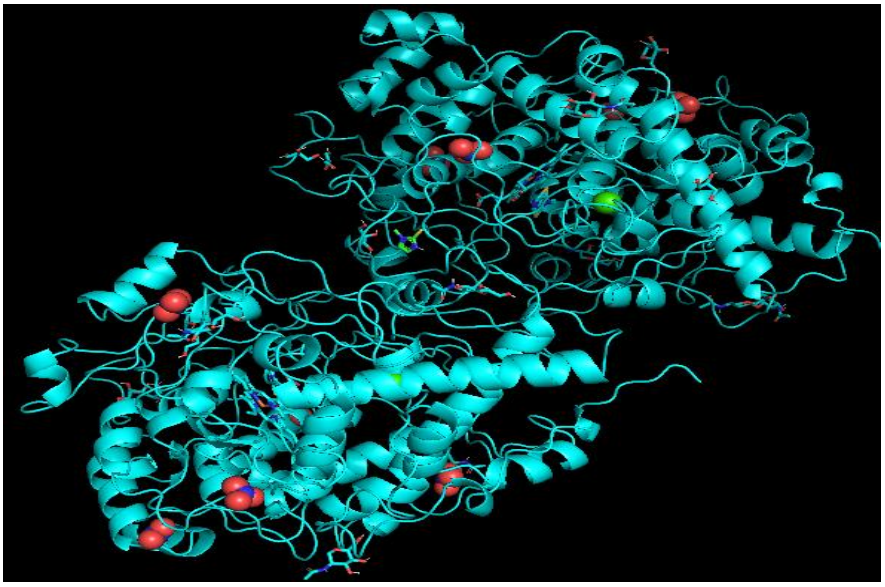

Figure 16: Vizualization of 7UUY and Methimazole in Pymol

5.5 7XW5 AND METHIMAZOLE

Receptor –7XW5 (TSH)
Ligand – Methimazole
Binding energy = -4.3 kcal/mol

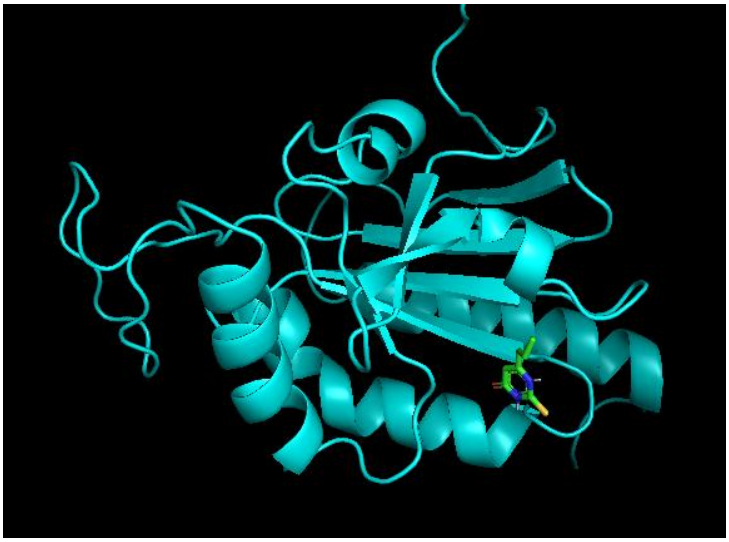

Figure 17: Vizualization of 7XW5 and Methimazole in Pymol

5.6 6SCJ and METHIMAZOLE

Receptor – 6SCJ (Thyroglobulin)
Ligand – Methimazole
Binding energy = -4.0 kcal/mol

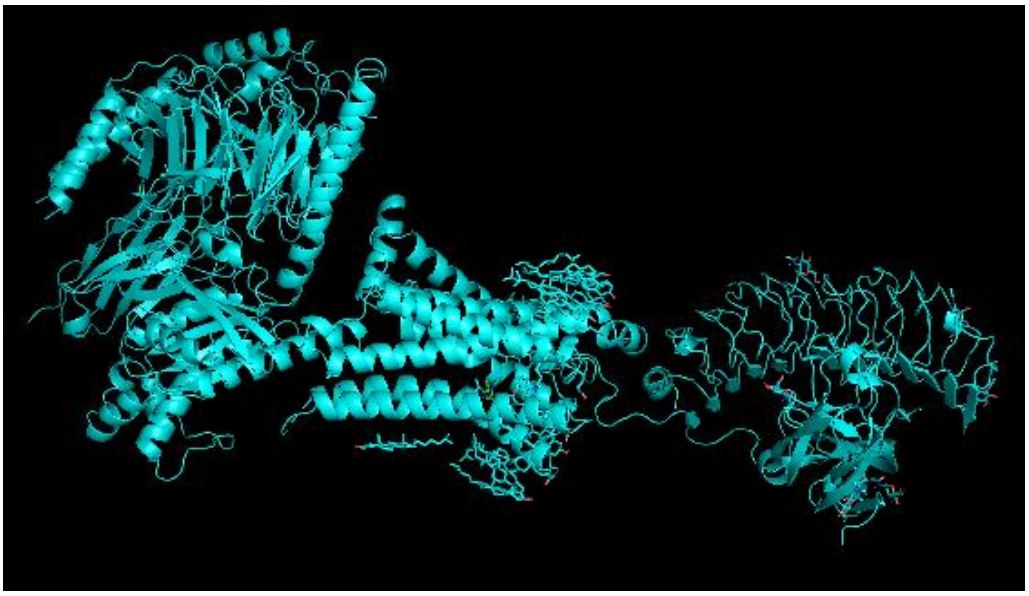

Figure 18: Vizualization of 6SCJ and Methimazole in Pymol

5.7 7UUY AND PROPYLTHIOURACIL

Receptor – 7UUY (Thyroid peroxidase)
Ligand –Propylthiouracil
Binding energy = -3.9kcal/mol

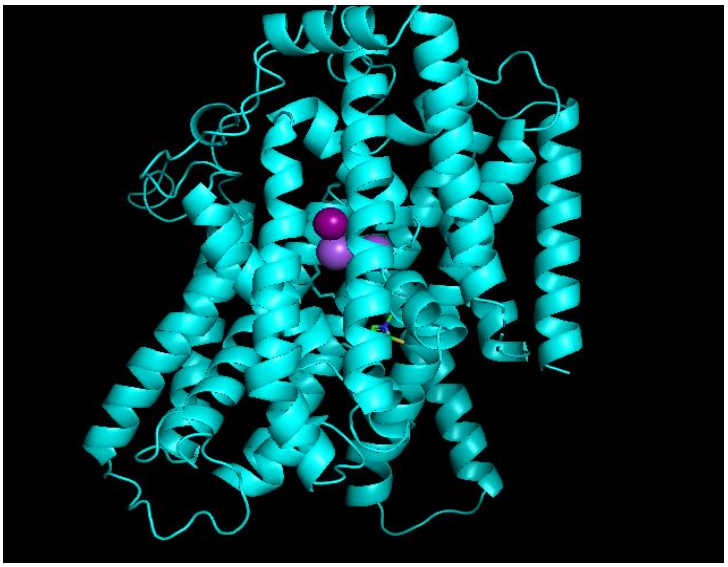

Figure 19: Vizualization of 7UUY and Propylthiouracil in Pymol

5.8 7UUY AND CARBIMAZOLE

Receptor – 7UUY (Thyroid peroxidase)
Ligand –Carbimazole
Binding energy = -3.2kcal/mol

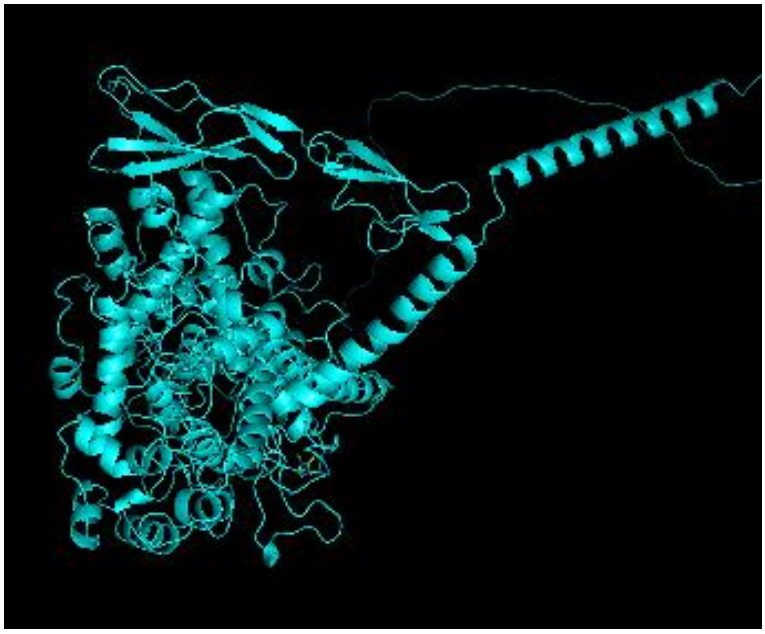

Figure 20: Vizualization of 7UUY and Carbimazole in Pymol

CHAPTER 5

CONCLUSION

Molecular docking is a bioinformatics-based theoretical simulation strategy. It is employed to study ligand and protein interaction profiles and predict their binding conformers and affinity through computational tools. By analysing all the phytochemical constituents identified in Lithospermum Officinale, we were able to determine their antithyroid and antioxidant properties and make predictions about how they would interact with thyroid drug targets. Utilising the docking software AutoDock vina, we relied on the ligands from Lithospermum officinale and docked it against receptors of several Graves'disease targets in our study to determine the best ligand-receptor interaction, which had a binding affinity almost as close to -8.6kcal/mol. All the molecules had negative Gibbs free energy that signifies good binding. In docking studies, the energy value between -4.0 to -8.0 kcal/mol is considered to be the best (according to the data given in Informatics in Medicine Unlocked Volume 5). The binding energy score for Thyroid peroxide (Graves disease) - Lithospermic acid was found to be the best in our study and it was -8.6 kcal/mol. The binding energy range of docked compounds of our research plant was from -6.9 to -8.6 kcal/mol. The binding energy score for various targets (Graves disease) - Methimazole, Carbimazole, Propylthiouracil the various compounds of the the current day drug for Graves' Disease was found to be the best in our study and it was -4.5 kcal/mol. The binding energy range of docked compounds of our research plant was from -3.2to -4.5kcal/mol. Based on the comparison between the binding affinity of current day drug and our phytochemical we concluded that the pythochemical of our plant is very efficient in curing the disease and its various after effects.

CHAPTER 6

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